BrandGuard AI

Intelligent Pre-Publication Brand Governance Platform

Round 3 — The Product (Final Evaluation) | Final Product Submission Document

Team Innovex | Team Leader: Ajay Kale | We Code Coders Hackathon | November 2026

> **EXECUTIVE JUDGE NOTE — ZERO-WALKTHROUGH READY:** A judge will open our link and use it on their own without anyone explaining anything. We have built 1-Click Instant Sign-In for all 4 roles directly on the login screen, an in-app 1-Click Role Switcher in the sidebar, and dual sample loaders (Violative vs. Compliant Green Pass) so anyone can evaluate the full system in seconds without typing.

1. Final Submission Links & Access Details

All services are fully deployed, interconnected, and live in production with global high availability:

Resource	URL / Access Point	Verification Status
Live Frontend Application	https://brandguard-frontend-6swq.onrender.com	Active (SPA Rewrite /* -> /index.html)
Backend REST API	https://brandguard-api-zfcd.onrender.com	Active (FastAPI + Auto-Lifespan Seeding)
Interactive API Docs	https://brandguard-api-zfcd.onrender.com/docs	Active (OpenAPI 3.1 Swagger UI)
Source Code Repository	https://github.com/AJJU3836/BrandGuard-AI	Active (Public GitHub Repository)
Final Product PDF	docs/BrandGuard_AI_Round_3_Submission.pdf	Delivered (Complete Round 3 Specification)

Demo Accounts: Pre-Seeded Enterprise Demo Accounts (Universal Password: password123):

Persona Role	Account Email	Default Password	Governance Role & Scope
Admin	kaleajay626@gmail.com	password123	Full administration, system health, workspace & audit logs

📊 Manager	mahesh@innovex.com	password123	Policy management, guideline-gap analytics, risk heatmaps
🔍 Reviewer	asma@innovex.com	password123	Draft approval, verification, request changes, findings audit
✍️ Creator	shrushti@innovex.com	password123	Draft creation, 1-click suggestion fixes, visual diff, image check

2. Complete List of Features Built (Full Product Inventory)

BrandGuard AI is an end-to-end pre-publication governance platform designed to prevent reputational, legal, and regulatory penalties. The complete platform encompasses:

2.1 Brand Kit Ingestion & Governance Engine

- **Multi-Facet Asset Representation:** Stores comprehensive brand rules: raw guideline prose, approved terminology, banned terms, approved corporate facts, tone rules, and primary hex color palettes.
- **Semantic Guideline Indexing:** Chunked into logical clauses and indexed using cosine TF-IDF vector similarity for sub-second relevant clause retrieval during review.
- **Workspace Multi-Tenancy:** Ensures strict tenant data boundary enforcement across disparate marketing units or brands.

2.2 Dual-Engine Real-Time Verification Pipeline

- **Deterministic Rule Engine:** High-speed deterministic checking (<50ms) for exact and regex banned phrases ('cheap', 'magic software', '#1 in India', '100% safe').
- **Claim Substantiation & Regulatory Check:** Checks factual claims against verified brand facts. Matches claims to ASCI (Advertising Standards Council of India) Chapter 1 truthfulness standards.
- **Tone Analysis Engine:** Detects aggressive, hyperbolic, or off-brand copy against configured voice principles.
- **Explainable AI Review Engine:** LLM grounded exclusively in retrieved guidelines to prevent hallucinations. Every finding provides an exact citation, severity, and suggested correction.
- **Dual-Engine Telemetry Badging:** Every flag is explicitly labeled as 'Rule Engine' or 'AI Evaluated' to guarantee transparent auditability.

2.3 Multimodal Vision & Accessibility Engine (Round 3 Improvement 1)

- **Color Palette Compliance:** Extracts dominant colors from creative images and validates them against the brand's approved palette with delta-E tolerance.
- **Alt-Text Audit:** Checks image alt-text presence and quality to ensure enterprise inclusivity.
- **WCAG 2.1 AA Contrast Ratio:** Calculates exact relative luminance and contrast ratios under WCAG 2.1 AA specifications (4.5:1 standard text, 3:1 graphical elements).
- **Visual Audit Interface:** Displays uploaded image preview, file weight, dimensions, and visual finding cards with 1-click category filters.

2.4 Interactive Visual Diff & 1-Click Suggestion Application (Round 3 Improvement 2)

- **Side-by-Side & Unified Diff:** Interactive word-by-word visual diff between draft iterations showing green additions (+word) and red strike-through deletions (~word~).
- **1-Click Suggestion Application:** Clicking 'Accept Suggestion' on any finding instantly replaces the violative phrase in the editor via regex token replacement.
- **Automated Version Control:** Every accepted suggestion creates an immutable DraftVersion snapshot and transitions finding status to 'accepted'.
- **Creator Custom Override:** Creators can click 'Edit Replacement' to fine-tune suggested copy before applying it.

2.5 Guideline-Gap Intelligence & Root-Cause Analytics Dashboard (Round 3 Improvement 3)

- **Governance Health Score:** Live calculation (0–100%) synthesizing violation severity, resolution rates, and drafting volume across the workspace.
- **Suggestion Acceptance Rate:** Tracks the percentage of AI and rule recommendations accepted by content creators.
- **Guideline-Gap Root-Cause Engine:** Clusters recurring violations to identify guidelines that creators consistently misunderstand.
- **Format Risk Heatmap Matrix:** Visual breakdown identifying which formats (Social Post, Email, Blog, Press Release) carry the highest governance exposure.

2.6 Workflow, RBAC & Audit Log

- **State Transition Machine:** Creator submits draft -> Reviewer / Manager reviews findings -> Request Changes or Approve.
- **RBAC Enforcement:** Granular role-based capability enforcement (Creators cannot self-approve; Reviewers/Managers can authorize publication).
- **Governance Audit Log:** Append-only tamper-evident ledger logging every action (creation, review, suggestion acceptance, status change).
- **Audit Report Exporter:** 1-click export of complete finding logs as formatted CSV or structured JSON.

3. Technical Architecture & Implementation Summary

BrandGuard AI utilizes a decoupled, high-resilience architecture engineered for instant feedback and fault tolerance:

3.1 System Architecture Topology

1. Client Layer: React 18 SPA compiled via Vite, served via Render Global CDN with SPA URL rewrites.
2. API Gateway: FastAPI asynchronous ASGI server running on Python 3.11 with Uvicorn.
3. Verification Pipeline: Parallel dispatch between Rule Engine (regex & lookup) and AI Engine (retrieval-grounded LLM).
4. Multimodal Vision Processor: Pillow-based image matrix luminance extractor and Gemini Vision API.
5. Storage & Persistence: Dual-strategy PostgreSQL 16 on Render with automated, zero-downtime SQLite fallback.

3.2 Multimodal WCAG Accessibility Algorithm

Accessibility contrast calculations follow the official W3C WCAG 2.1 specifications. For any sRGB color component C (R, G, or B scaled to 0-1):

If C <= 0.04045: C_lin = C / 12.92
Else: C_lin = ((C + 0.055) / 1.055) ^ 2.4
Relative Luminance (L): L = 0.2126 * R_lin + 0.7152 * G_lin + 0.0722 * B_lin

Contrast Ratio (CR): CR = (L1 + 0.05) / (L2 + 0.05), where L1 >= L2.
Thresholds: Normal text passes if CR >= 4.5:1; large text/visual elements pass if CR >= 3.0:1.

3.3 Technology Stack Inventory

Layer	Technology	Production Purpose
Frontend Framework	React 18 + Vite	High-performance SPA with sub-second HMR and optimized chunking
Styling & Icons	Tailwind CSS + Lucide React	Responsive UI, accessible color contrast, and iconography
Diff Engine	diff (npm)	Word-by-word diff calculation and highlight token generation
Backend API	FastAPI (Python 3.11)	High-throughput asynchronous ASGI web server with OpenAPI 3.1 docs
ORM & Data Modeling	SQLAlchemy 2.0 + Pydantic v2	Type-safe relational mapping and strict runtime schema validation
Dual Database	PostgreSQL 16 + SQLite Fallback	Production PostgreSQL with automated 5s timeout fallback to SQLite
Vision & Processing	Pillow (PIL) + Gemini Vision	Color matrix quantization, luminance extraction, and visual checks
AI Reasoning	Gemini API (Grounded)	Explainable compliance critique grounded in brand kit clauses
Deployment Host	Render Cloud Platform	Static CDN for frontend, Containerized Web Service for backend API

4. Improvements Completed After Round 2: The Three Delivered Commitments

In Section 5 of our Round 2 Submission Document, Team Innovex committed to three specific improvements for Round 3. We have engineered and delivered exactly those three improvements:

Improvement Commitment	Round 2 Promise	Round 3 Implementation	Verification Status
Improvement 1: Gemini Vision & WCAG Engine	Image review, palette hex check, alt-text presence, WCAG 2.1 AA contrast ratio.	Base64 Gemini Vision integration, PIL relative luminance math, alt-text audit, visual card UI.	☑ 100% Delivered & Live (POST .../review/image)
Improvement 2: Interactive Visual Diff & 1-Click Fix	Side-by-side diff between draft versions, inline accept/apply suggestion controls, version timeline.	Regex token replace, automatic DraftVersion snapshotting, 'Accept Suggestion' button, split/unified diff view.	☑ 100% Delivered & Live (POST .../findings/{id}/apply)
Improvement 3: Guideline-Gap Intelligence Dashboard	Violation aggregation, guideline health score, content type breakdown, gap recommendations.	Governance Health Score (0-100%), Content Format Risk Heatmap, Guideline-Gap diagnostics with 1-click update links.	☑ 100% Delivered & Live (GET .../dashboard/stats)

4.1 Deep Dive: Improvement 1 — Gemini Vision Multimodal & Accessibility Engine

- **The Problem Solved:** Prior to Round 3, BrandGuard AI was text-centric. Off-brand visual assets, unapproved hex palettes, and inaccessible graphics went undetected.
- **Engine Implementation:** Endpoint POST /api/workspaces/{ws}/drafts/{draft_id}/review/image processes multipart uploads. PIL computes relative luminance of dominant colors. Gemini Vision verifies brand visual coherence and alt-text appropriateness.
- **User Experience:** The review workspace surfaces an 'Audited Visual Asset' card with image preview, alt-text status, and dedicated 1-click filters ('Visual Rules', 'WCAG Accessibility').

4.2 Deep Dive: Improvement 2 — Interactive Visual Diff & 1-Click Suggestion Application

- **The Problem Solved:** Previously, creators had to read suggestions and manually edit text in a separate screen, risking typos and human error.
- **Engine Implementation:** Endpoint POST /api/workspaces/{ws}/drafts/{draft_id}/review/findings/{id}/apply extracts suggested replacements using pattern matching, performs atomic replacement, increments draft version number, saves DraftVersion record, and logs an AuditLog event.
- **User Experience:** Every finding has a prominent 'Accept Suggestion' button. In-editor visual diff allows creators to inspect changes word-by-word (green addition, red strikethrough) with a 1-click toggle between Split View and Unified View.

4.3 Deep Dive: Improvement 3 — Guideline-Gap Intelligence & Root-Cause Analytics Dashboard

- **The Problem Solved:** Managers previously only saw raw violation counts without actionable understanding of why guidelines were failing.
- **Engine Implementation:** Endpoint GET /api/workspaces/{ws}/dashboard/stats aggregates historical audit data. Calculates Governance Health Score, Suggestion Acceptance Rate, and analyzes top recurring guideline breaches.

- **User Experience:** Dashboard features a real-time Health Gauge, a Content Format Risk Heatmap, and 'Guideline Gaps & Recommendations' cards equipped with direct 1-click links to update the Brand Kit.

5. Product Reliability & Zero-Walkthrough Hardening (Self-Sustaining Product)

A core directive of Round 3 is: 'A judge will open your link and use it on their own, with nobody explaining anything. Nothing should break or hang.' To ensure complete independence, we implemented:

- **1-Click Instant Demo Authentication:** Universal demo password 'password123' clearly displayed on the login card. 4 pre-seeded personas (Admin, Manager, Reviewer, Creator) equipped with 1-Click Instant Sign-In.
- **Sidebar Live Role Switcher:** Judges can switch personas (e.g., from Creator to Reviewer to Manager) directly from the sidebar in 1 click without logging out or retyping.
- **Dual 1-Click Sample Generators:** CreateDraft page includes 'Load Violative Sample (Demo)' to test violation detection and 'Load Compliant Sample (Green Pass)' to test clean 100% approval passes.
- **Deep-Link & Refresh Resiliency:** Configured Render static rewrite '/* -> /index.html'. Evaluators can refresh on any deep URL (/drafts/:id, /brand-kits) without 404 errors.
- **Cold-Start Pre-Warm & Visual Indicator:** Frontend pings the backend on initial load. If the server is on a cold-start sleep (~25s on Render free tier), a friendly pulsating alert explains the wakeup process.
- **Dual Database Fallback:** PostgreSQL connection uses a 5-second socket timeout; if the remote connection drops or times out, the backend seamlessly fails over to local SQLite.
- **Double-Click & Re-Submission Guards:** All primary action buttons (Submit Draft, Accept Suggestion, Re-Run Audit, Approve) are debounced and disabled during asynchronous processing.

6. Comprehensive Testing Details & Verification Suite

BrandGuard AI has been validated through rigorous automated and manual test harnesses:

6.1 Automated Integration & Regression Suite

Our test suite (scripts/test_e2e_polish.py) executes end-to-end against live APIs using FastAPI TestClient:
• Test 1 (Auth & Multi-Persona): Login for Admin, Manager, Reviewer, and Creator verified (200 OK).
• Test 2 (Self-Healing Registration): Newly registered users auto-link to default workspace (Verified: workspace_id != None).
• Test 3 (Violative Audit): Draft with unapproved claims audited -> 4 violations detected across Rule & AI engines.
• Test 4 (1-Click Suggestion Application): Finding accepted -> Draft updated to v2, finding status marked 'accepted'.
• Test 5 (Compliant Green Pass): Compliant draft audited -> Exactly 0 open violations, 100% health pass confirmed.
• Test 6 (Dashboard Telemetry): Health score, acceptance rate, and guideline gaps verified.
Result: 100% PASS across all automated test suites.

6.2 Role-Based Access Control (RBAC) Test Matrix

Role Persona	Create Draft	Accept Suggestion	Approve / Request Changes	Manage Brand Kit
✍️ Creator (Shrushti)	☑ Allowed	☑ Allowed	❌ Forbidden (Hidden)	❌ Read Only

🔍 Reviewer (Asma)	☑ Allowed	☑ Allowed	☑ Authorized	❌ Read Only
👔 Manager (Mahesh)	☑ Allowed	☑ Allowed	☑ Authorized	☑ Full Edit
👑 Admin (Ajay)	☑ Allowed	☑ Allowed	☑ Authorized	☑ Full Control

7. Usability & Real-World Fit (Market Validation)

BrandGuard AI directly solves critical compliance liabilities for modern marketing organizations:

- **ASCI Regulatory Shield:** According to ASCI Annual Reports, over 98% of scrutinized advertisements violated substantiation guidelines in FY 2025-26. BrandGuard AI catches unsubstantiated claims pre-publication.
- **Agency & Enterprise Workflow:** Marketing teams, external agencies, and freelancers submit drafts to a single governance gateway, eliminating manual review bottlenecks.
- **The 'AI Suggests, Humans Decide' Principle:** BrandGuard AI does not auto-publish or silently modify copy. It highlights issues, suggests approved rewrites, and leaves final approval in the hands of authorized human managers.

8. Product Roadmap & Future Development Plan

BrandGuard AI is structured as a scalable commercial SaaS platform with a clearly phased development roadmap:

Horizon / Phase	Feature Deliverables	Target Value & Market Impact
Phase 1 (Q1 2027)	Figma & Canva Plugins	Audit vector assets, layouts, and typography directly inside design software before exporting.
Phase 2 (Q2 2027)	Chrome Extension for CMS & Social	Real-time compliance linter inside WordPress, LinkedIn, Twitter, and Hubspot drafting interfaces.
Phase 3 (Q3 2027)	Slack & Teams Governance Bot	Instant draft reviews via @BrandGuard mention in team communication channels.
Phase 4 (Q4 2027)	Enterprise Multi-Tenant SaaS & SSO	SAML/Okta enterprise authentication, custom AI model fine-tuning per brand, and localized legal packs.

9. Team Innovex & Project Contributions

Team Member	Role	Round 3 Core Contributions
Ajay Kale	Team Leader / Backend & AI Lead	Multimodal Gemini Vision API, WCAG contrast mathematics, 1-click suggestion endpoint, PostgreSQL/SQLite fallback, Render deployment, E2E test suite.
Mahesh Gadekar	Frontend & UI/UX Lead	Interactive visual diff UI, side-by-side version comparison, visual asset audit card, demo role switcher, 1-click sample loaders.
Asma Inamdar	Research & Governance Lead	ASCI guideline rule sets, WCAG 2.1 AA accessibility standards verification, multi-persona RBAC testing, error state validation.
Shrushti Kharadkar	Documentation & Quality Assurance	Round 3 Final Product Document, README clone-and-run verification, cross-browser compatibility, test scenario auditing.